

Quiz 5. Let D be the set of all strings whose length is a positive multiple of 128 bits. Define $\mathcal{T}: \{0, 1\}^{128} \times D \rightarrow \{0, 1\}^{128}$ as follows for any $K \in \{0, 1\}^{128}$, $M \in D$:

Algorithm $\mathcal{T}_K(M)$:
 $T' \leftarrow \text{CBC-MAC}_K(M)$
 $T \leftarrow \text{AES}_K(T')$
 Return T

Above, CBC-MAC_K denotes CBC-MAC using AES as the underlying block cipher with key K .

Part A. (2 points.) What is the difference between $\mathcal{T}$ and ECBC-MAC?

Part B. (8 points.) Show that $\mathcal{T}$ is not a secure MAC. Specify an adversary in succinct pseudocode and briefly justify its advantage and running time. Your adversary must *not* break AES as a PRF (no exhaustive key search, birthday attacks, *etc.*).